

PDF Extraction Quality Test

Validating PyMuPDF+OCR output for LLM fiscal shock detection

Fiscal Shocks Project

June 30, 2026

Overview

This notebook tests whether the local PyMuPDF PDF extraction (with OCR for scanned documents) produces text that Claude 3.5 Sonnet can effectively use for:

- **Model A:** Act detection (identify fiscal legislation passages)
- **Model B:** Motivation classification (spending-driven, countercyclical, etc.)
- **Model C:** Information extraction (timing, magnitudes from tables)

Setup

```
library(tidyverse)
library(here)
here::i_am("notebooks/test_text_extraction.qmd")

# Source the local extraction function
source(here("R/pull_text_local.R"))

# Load ground truth data
us_shocks <- read_csv(here("data/raw/us_shocks.csv"))
us_labels <- read_csv(here("data/raw/us_labels.csv"))
```

Phase 1: Sample Selection

We select documents with well-known fiscal acts that have clear ground truth labels.

```
# Key test cases with known acts
test_cases <- tribble(
  ~act_name, ~year, ~motivation, ~expected_url_pattern,
  "Revenue Act of 1964", 1964, "Long-run", "ERP.*1965",
  "Tax Reform Act of 1986", 1986, "Long-run", "ERP.*1987",
  "Economic Recovery Tax Act of 1981", 1981, "Long-run",
  "ERP.*1982",
  "Tax Reduction Act of 1975", 1975, "Countercyclical",
```

```
"ERP.*1976",
  "Revenue Act of 1950", 1950, "Spending-driven", "ERP.*1951"
)

test_cases
```

```
# A tibble: 5 × 4
  act_name year motivation
expected_url_pattern
<chr> <dbl> <chr> <chr>
1 Revenue Act of 1964 1964 Long-run
ERP.*1965
2 Tax Reform Act of 1986 1986 Long-run
ERP.*1987
3 Economic Recovery Tax Act of 1981 1981 Long-run
ERP.*1982
4 Tax Reduction Act of 1975 1975 Countercyclical
ERP.*1976
5 Revenue Act of 1950 1950 Spending-driven
ERP.*1951
```

```
# Get actual URLs for test documents
# Using Fraser St. Louis Fed ERP archive
test_urls <- c(
  "https://fraser.stlouisfed.org/files/docs/publications/ERP/
1965/ERP_1965.pdf",
  "https://fraser.stlouisfed.org/files/docs/publications/ERP/
1982/ERP_1982.pdf"
)

cat("Test URLs:\n")
```

Test URLs:

```
cat(test_urls, sep = "\n")
```

```
https://fraser.stlouisfed.org/files/docs/publications/ERP/1965/
ERP_1965.pdf
https://fraser.stlouisfed.org/files/docs/publications/ERP/1982/
ERP_1982.pdf
```

Phase 2: Extract Sample Documents

```
# Extract text from sample PDFs using local PyMuPDF+OCR
# This may take 5-15 minutes for scanned documents (OCR needed)
message("Starting local extraction...")

sample_results <- pull_text_local(
  pdf_url = test_urls,
  output_dir = here("data/extracted"),
  workers = 4,
  ocr_dpi = 200
)

message("Extraction complete!")
```

Extraction Summary

```
sample_results |>
  mutate(
    doc_name = basename(test_urls)
  ) |>
  select(doc_name, n_pages, ocr_used, extraction_time,
         extracted_at)
```

```
# A tibble: 2 × 5
  doc_name      n_pages ocr_used extraction_time extracted_at
<chr>          <int> <lgl>          <dbl> <dtm>
1 ERP_1965.pdf     300 TRUE          277. 2026-01-17 15:35:36
2 ERP_1982.pdf     368 TRUE          368. 2026-01-17 15:35:36
```

Phase 3: Text Quality Inspection

3.1 Basic Text Readability

```
# Show sample text from first document
cat("=== Sample text from ERP 1965 (first 2000 chars) ===\n\n")
```

```
=== Sample text from ERP 1965 (first 2000 chars) ===
```

```
first_doc_text <- unlist(sample_results$text[[1]])
cat(substr(paste(first_doc_text, collapse = "\n\n"), 1, 2000))
```

```
_ Jaturday 1960
|
Together With
```

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of the
COUNCIL OF ECONOMIC ADVISERS
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<http://fraser.stlouisfed.org>,
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January 1965
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UNITED STATES GOVERNMENT PRINTING OFFICE
WASHINGTON: 1965
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```

3.2 Check for Known Act Names

```

# Acts we expect to find in these documents
expected_acts <- c(
  "Revenue Act of 1964",
  "Economic Recovery Tax Act",
  "Tax Equity and Fiscal Responsibility Act"
)

# Search for act names in extracted text
all_text <- paste(unlist(sample_results$text), collapse = " ")

act_detection <- map_dfr(expected_acts, function(act) {
  found <- str_detect(all_text, fixed(act))
  tibble(
    act_name = act,
    found = found,
    mentions = sum(str_count(all_text, fixed(act)))
  )
})

act_detection

```

```

# A tibble: 3 × 3
  act_name                                found mentions
  <chr>                                <lgl>     <int>

```

1 Revenue Act of 1964	TRUE	4
2 Economic Recovery Tax Act	TRUE	17
3 Tax Equity and Fiscal Responsibility Act	FALSE	0

```
# Act detection rate
detection_rate <- mean(act_detection$found)
cat(sprintf("Act detection rate: %.0f%% (%d/%d)\n",
            detection_rate * 100,
            sum(act_detection$found),
            nrow(act_detection)))
```

Act detection rate: 67% (2/3)

```
if (detection_rate < 0.8) {
  warning("Act detection rate below 80% - may need to check
  extraction quality")
}
```

3.3 Numeric Value Preservation

```
# Check for dollar amounts (critical for Model C)
dollar_pattern <- "\\$\\s*\\d+\\.?\\d*\\s*(billion|million|B|M)"
year_pattern <- "\\b(19[4-9]\\d|20[0-2]\\d)\\b"

numeric_metrics <- tibble(
  metric = c("Dollar amounts", "Year mentions", "Percentage
  values"),
  pattern = c(dollar_pattern, year_pattern, "\\d+\\.?\\d*\\s*percent"),
  count = c(
    sum(str_count(all_text, regex(dollar_pattern, ignore_case =
    TRUE))),
    sum(str_count(all_text, year_pattern)),
    sum(str_count(all_text, regex("\\d+\\.?\\d*\\s*percent",
    ignore_case = TRUE)))
  )
)

numeric_metrics
```

```
# A tibble: 3 × 3
  metric      pattern      count
  <chr>      <chr>      <int>
1 Dollar amounts "\\$\\s*\\d+\\.?\\d*\\s*(billion|million|B|
```

M)"	214	
2 Year mentions	"\\b(19[4-9]\\d 20[0-2]\\d)\\b"	6201
3 Percentage values	"\\d+\\.?\\d*\\s*percent"	527

3.4 Fiscal Policy Terms

```
# Check for key fiscal policy terminology
fiscal_terms <- c(
  "tax cut", "tax reduction", "fiscal policy",
  "federal budget", "deficit", "expenditure",
  "revenue", "appropriation"
)

term_counts <- map_dfr(fiscal_terms, function(term) {
  tibble(
    term = term,
    count = sum(str_count(all_text, regex(term, ignore_case =
TRUE)))
  )
})

term_counts |>
  filter(count > 0) |>
  arrange(desc(count))
```

```
# A tibble: 8 × 2
  term          count
<chr>         <int>
1 expenditure    246
2 deficit        192
3 revenue         74
4 tax cut         50
5 federal budget  32
6 fiscal policy   31
7 tax reduction   17
8 appropriation    4
```

Phase 4: LLM Readiness Tests

4.1 Passage Length Analysis

```
# Check if pages are reasonable length for LLM context
first_doc_pages <- sample_results$text[[1]]
page_lengths <- map_int(first_doc_pages, nchar)

tibble(
```



```
metric = c("Min page length", "Max page length", "Mean page
length", "Total chars"),
  value = c(min(page_lengths), max(page_lengths),
mean(page_lengths), sum(page_lengths))
) |>
mutate(value = scales::comma(value))
```

```
# A tibble: 4 × 2
  metric      value
  <chr>      <chr>
1 Min page length 85
2 Max page length 7,831
3 Mean page length 3,158
4 Total chars    947,522
```

4.2 Token Estimation

```
# Rough token estimate (1 token ~ 4 chars for English)
total_chars <- sum(map_int(unlist(sample_results$text), nchar))
estimated_tokens <- total_chars / 4

cat(sprintf("Estimated                                tokens:
%s\n", scales::comma(estimated_tokens)))
```

```
Estimated tokens: 520,314
```

```
cat(sprintf("Claude 3.5 Sonnet context: 200K tokens\n"))
```

```
Claude 3.5 Sonnet context: 200K tokens
```

```
cat(sprintf("Fits in context: %s\n",
  ifelse(estimated_tokens < 200000, "YES", "NO -
need chunking")))
```

```
Fits in context: NO - need chunking
```

4.3 Sample Passage for Model A Test

```
# Find a passage mentioning a fiscal act
first_doc_pages <- sample_results$text[[1]]

act_passages <- map_dfr(seq_along(first_doc_pages), function(i) {
  page_text <- first_doc_pages[[i]]
```

```

if (str_detect(page_text, regex("act of \\d{4}", ignore_case =
TRUE))) {
  tibble(
    page = i,
    text = str_trunc(page_text, 500),
    has_act_mention = TRUE
  )
} else {
  NULL
}
})

if (nrow(act_passages) > 0) {
  cat("=== Sample passage with act mention (for Model A) ===\n\n")
  cat(act_passages$text[1])
} else {
  cat("No passages with 'Act of YYYY' pattern found")
}

```

```

=== Sample passage with act mention (for Model A) ===

```

or a lack of confidence in the dollar.
 Since 1946, therefore, we have
 come to recognize that the mandate of the Employment Act implies a
 series of objectives closely related to the goal of full
 employment:
 -rapid growth,
 - price stability, and
 -equilibrium in our balance of payments.
 Rapp GRowTH
 True prosperity means more than the full use of the productive
 powers available at any given time.
 It also means the rapid expansion
 of those powers.
 In the long run, it is only a growth of over-all ...

Phase 5: Quality Metrics Summary

```

# Compile all quality metrics
total_pages <- sum(sample_results$n_pages)
total_tables <- 0 # PyMuPDF doesn't extract tables separately

# Convert all values to character to avoid type mismatch in tribble
quality_report <- tribble(
  ~metric, ~value, ~target, ~status,
  "Documents extracted", as.character(nrow(sample_results)), "2",

```

```

"PASS",
  "Pages extracted", as.character(total_pages), ">0",
    ifelse(total_pages > 0, "PASS", "FAIL"),
  "OCR used", as.character(sum(sample_results$ocr_used)), "as
needed", "INFO",
  "Act name recall", sprintf("%.0f%%", detection_rate * 100),
">80%",
    ifelse(detection_rate >= 0.8, "PASS", "FAIL"),
  "Dollar amounts found", as.character(numeric_metrics$count[1]),
">0",
    ifelse(numeric_metrics$count[1] > 0, "PASS", "WARN"),
  "Year mentions", as.character(numeric_metrics$count[2]), ">10",
    ifelse(numeric_metrics$count[2] > 10, "PASS", "WARN"),
  "Fits in LLM context", ifelse(estimated_tokens < 200000, "YES",
"NO"), "YES",
    ifelse(estimated_tokens < 200000, "PASS", "WARN")
)

quality_report |>
  gt::gt()

```

metric	value	target	status
Documents extracted	2	2	PASS
Pages extracted	668	>0	PASS
OCR used	2	as needed	INFO
Act name recall	67%	>80%	FAIL
Dollar amounts found	214	>0	PASS
Year mentions	6201	>10	PASS
Fits in LLM context	NO	YES	WARN

Conclusions

```

pass_count <- sum(quality_report$status == "PASS")
warn_count <- sum(quality_report$status == "WARN")
fail_count <- sum(quality_report$status == "FAIL")
info_count <- sum(quality_report$status == "INFO")

cat(sprintf("\n=== QUALITY ASSESSMENT ===\n"))

```

```

=== QUALITY ASSESSMENT ===

```

```
cat(sprintf("PASS: %d | WARN: %d | FAIL: %d | INFO: %d\n\n",
            pass_count, warn_count, fail_count, info_count))
```

```
PASS: 4 | WARN: 1 | FAIL: 1 | INFO: 1
```

```
if (fail_count == 0) {
  cat("Extraction quality is SUFFICIENT for LLM processing\n")
  cat("  Proceed with full pipeline: tar_make(us_text)\n")
} else {
  cat("Extraction quality needs IMPROVEMENT before full run\n")
  cat("  Review failed metrics and adjust extraction settings\n")
}
```

```
Extraction quality needs IMPROVEMENT before full run
  Review failed metrics and adjust extraction settings
```

Next Steps

Based on the quality assessment:

1. **If PASS:** Run `tar_make(us_text)` to extract all 350 documents
2. **If WARN:** Review warnings but proceed with caution
3. **If FAIL:** Debug extraction issues before full run

Recommended Actions

```
if (fail_count > 0) {
  cat("Issues to address:\n")
  quality_report |>
    filter(status == "FAIL") |>
    pull(metric) %>%
    paste("-", .) |>
    cat(sep = "\n")
}
```

```
Issues to address:
- Act name recall
```

Extraction Performance

```
# Summary of extraction times
cat("=== Extraction Performance ===\n")
```

```
=== Extraction Performance ===
```

```
cat(sprintf("Total documents: %d\n", nrow(sample_results)))
```

Total documents: 2

```
cat(sprintf("Total pages: %d\n", sum(sample_results$n_pages)))
```

Total pages: 668

```
cat(sprintf("Total extraction time: %.1f seconds\n",  
            sum(sample_results$extraction_time, na.rm = TRUE)))
```

Total extraction time: 644.2 seconds

```
cat(sprintf("Average time per document: %.1f seconds\n",  
            mean(sample_results$extraction_time, na.rm = TRUE)))
```

Average time per document: 322.1 seconds

```
cat(sprintf("Average time per page: %.2f seconds\n",  
            sum(sample_results$extraction_time, na.rm = TRUE) /  
            sum(sample_results$n_pages)))
```

Average time per page: 0.96 seconds